

Ethical Case Study of the Chernobyl Nuclear Disaster

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1. Introduction

During a fateful day in April 1986, reactor 4 of Chernobyl Nuclear Power Plant in Pripjat, Ukrainian SSR, Soviet Union exploded creating the worst nuclear disaster in recorded history. This happened during a safety test releasing 400 times more radiation than the bomb dropped in Hiroshima. It killed the plant operators and first responders while at the same time forcing the mass evacuation of hundreds of thousands of people in the vicinity. Even to this day Chernobyl remains having a 30 km exclusion zone to prevent the spread of contaminants.

This case study will examine the decision making that led to the convergent issues creating the Chernobyl disaster situation. More specifically I will be looking at the safety tests on April 26 and the ethical decision making that led to the disaster on that day, along with the engineering ethical issues in the design of the RBMK reactor. I will be analyzing using the Patient Centered Deontology and Reflexive principlsm frameworks. This approach streamlines the conclusion of moral right or wrong through the use of rights and duties. As such, for the case of Chernobyl we can extend the deontological framework to encompass almost all decisions taken given how much impact those decisions had to the general population at the time but also how much impact it has to us today in regards to how the events affected policies now. In the same lense we can apply a Reflexive principlist edge to our analysis as a way of streamlining ethical decision making by focusing on applying the four principles: Beneficience, Nonmaleficence, Justice and Respect for autonomy.

Aside from the direct consequences of the explosion and fall out, the Chernobyl disaster caused a lot of changes in engineering decision making, dangerous technology policy and public perception of new technology. These three points are very defined when we look at similar upcoming technological advancements in our era like artificial intelligence. Understanding and analyzing the ethical decision making mistakes that occurred during the Chernobyl disaster will ultimately be incredibly helpful in navigating the modern landscape of technological advancement.

2. Background

To begin the background picture, we will first need to talk about the design of the reactor itself. The RBMK-1000 (Reaktor Bolshoy Moshchnosti Kanalnyy, or High-Power Channel-type Reactor) was a Soviet-designed graphite-moderated, water-cooled reactor. Unlike the light-water reactor designs, the RBMK had a positive void coefficient at low power levels. This meant that as water turned into steam the nuclear reaction accelerated instead of slowing down causing possible positive feedback loops leading to inherently dangerous reactor environments which is the first ethical issue. On top of this issue the control rods that were meant to create a balance and reduce reactivity in the reactor had graphite tips instead of boron which for a short period increase the reactivity in the reactor giving us the second ethical issue. So we have two ethical issues that when combined together create an extremely unstable positive feedback loop system with a potential excitor in the control rods. Furthermore, it was known by the design engineers that this reactor design had this positive feedback environment inherently baked into the way it operates and did nothing, giving us our third ethical issue in the design area.

Next I will talk about what exactly happened during the April 26 safety testing leading up to the disaster. The explosion occurred during a test that was designed to examine whether or not during a power outage the turbines could generate enough electric power to power the emergency core cooling system in the 1 minute “redundancy” gap that the plant designers added before the backup gas generators went up. On the night that the safety test was scheduled to be done the plant director Viktor Bryukhanov and chief engineer Nikolai Fomin authorized the testing procedure under heavy pressure by the Soviet grid demand schedule, this is the fourth ethical flaw that led to the disaster. Because the test was administered in unsafe conditions under career pressure of the engineers and technicians, they made the mistake of when the reactor was in the low power mode with the disabled cooling system, the plant operators proceed with the test and within the seconds of the water rushing into the positive void coefficient reactor with the graphite tips it created the explosion which still affects the world to this day.

2.1 Stakeholders

This case study involves a very wide range of stakeholders given the gravity of the situation however the main stakeholders being:

- Reactor Designers
- Plant Operators
- Soviet Leadership
- General Public of Ukraine near Chernobyl and in hindsight the general public of Europe.
- Future Generations

3. Method: Ethical Frameworks

So far we have learned many different types of ethical frameworks, however I believe the two most fitting frameworks for this case study are Deontology and Reflexive Principlism. Deontology fits in this analysis due to the fact that the ethical issues found in the Chernobyl case all seem to stem from the misuse of stakeholder duties and responsibilities. Because of that, we can use deontology as a tool to sharpen our ethical decision making patterns much more than with something like consequentialism which often has an open ended future knowledge requirement. To be more specific in this case I want to look at Agent and Patient centered deontological frameworks, mainly due to the fact that these ethical issues arise from the primary agents being afraid for their career and family if they did not go along to get along during the USSR days. Famously there are a lot of people who inexplicably get the urge to jump off a balcony so from this point of view it makes sense that an Agent Centered Deontological framework would be good to analyze the decision making process of the primary agents themselves. Looking from hindsight though we can see that because of these tiny ethical mistakes being made at every step of the way, compounding into what we know know as one of the worst disasters in human history it would also be fair to take a patient centered deontological due to the scale of the pain suffering and loss that occurred and that could have been prevented as a result of the ethical mistakes made. For the case of Chernobyl we can also extend the deontological framework to encompass almost all decisions taken given how much impact those decisions had to the general population at the time but also how much impact it has to us today in regards to how the events affected policies, responsibilities and duties now. In this day and age now with the rise of AI and other very impactful technologies to our daily lives we as engineers need to look back at examples of history and calibrate our ethical decision making compass. This

in today's era could be done through a more Contractual Deontological framework however the more useful base fundamental analysis that I believe remains to be Agent/Patient Deontology. We can also look at this case through a Reflexive Principalist approach as there were a lot of decisions being made that could have been changed if the actor was using a reflexive principality approach leading to better overall outcomes for the general public.

3.1 Agent and Patient-Centered Deontology

Fundamentally, Deontology is a framework that ascribes moral correctness based on the rights and duties attributed to an actor. There are many different ways of thinking about rights and duties with the main two revolving around the actor and the patient or victim of an act. As an engineer it is your duty to provide certain safety measures in designs, it is also your duty to prioritize your family in certain situations. For an ethical analysis of the situation it seems to me that given the state of the Soviet Union at the time we need to combine both agent and patient centered deontological frameworks in order to understand the mental situation and decision making of the stakeholders that had agency in the situation.

We can think about Deontology and more specifically Agent and Patient Centered Deontology as frameworks that decide on the moral weight right or wrong of an action based on the rights and duties that that actor has. And that is also precisely how we would differentiate between an Agent vs Patient centered view which is through either the rights patients have or through the duties that the actor has.

3.2 Reflexive Principlism

Reflexive principlism is a framework of ethical decision making by focusing on maximizing four principles: Beneficence, Nonmaleficence, Justice, Respect for autonomy. What this does is that it gives the actor a lot of good tools to use when decision making. It combines a lot of good points from deontological consequentialist and virtue ethics frameworks into a practical and relatively simple way of determining and committing to a decision. Within reflexive principlism, we can take a look at beneficence as creating benefits for others and benefit maximization, nonmaleficence as harm minimization then Justice and Autonomy being the search for fair and equitable outcomes along with giving others agency to make their own decisions respectively. From these four tenets we can see that there are shades of consequentialism, deontology and virtue ethics in all key parts of this framework.

4. Analysis

To analyze the main ethical issue with this whole situation I believe that we only need to look at 2 main issues that would keep this case study from being too long. The reactor design and the control rod design. From an engineering perspective this is what led to the disaster scenario possibility in the first place. As engineers we are trained that the user needs to be assumed to be as unintelligent as possible and that whatever can go wrong will go wrong. In terms of creating the conditions for a perfect storm of destruction, the main ethical mistakes we need to look at are those two.

4.1 Ethical Issue 1: Positive Void Coefficient Reactor Design

Ethical Issue

Engineers had previous knowledge of positive void coefficients design flaw. They neglected to report this to the higher authorities nor did they try to find a solution around it similar to western reactors at the time which were negative void coefficient designs. It seems like the main issue relating to why they did not continue with the negative void design was due to budget constraints mixed together with pressure from higher ups and KGB to get things done.

Deontological Analysis

From an Agent centered point of view, it makes sense that they would keep on with the positive void coefficient reactor design. The engineers had first and foremost a duty to their family to keep on working and shutting up, with how easy it was for them to not only lose their career but their whole life over things like this I don't think that it is unethical for the engineers to simply use the positive void coefficient design. It is however an issue that they did not attempt to give more information to the operators in any way shape or form that the reactors had a positive void coefficient which could lead to a possible fallout scenario.

From a Patient centered point of view the general public had a right to give them power in their homes safely. In the lens of rights to the general public stakeholders, this was a spectacular failure on every part. Not only did they not try to fight for the rights of the stakeholders but they did not implement good enough failsafes, emergency protocols or any type of guidance in the known positive feedback loop instability case.

Reflexive Principlism Analysis

From a reflexive principalism view we can take a look at the four points. From a beneficence perspective the design engineers seem to be at least giving a lot of power which the stakeholders want which helps out a lot of the general public. From a nonmaleficence view this is a horrible design decision as it is actively leading to chaotic reactor situations that lead to positive feedback loops and explosions. From a Justice perspective it seems to me that it is equitable decision, everyone gets affected the same way from radiation poisoning, and finally from an Autonomy point of view it does not bode well given that none of the stakeholders have any say in the decision making process of the engineers and designers. Because of this lack of autonomy non maleficence and neutrality on justice we can clearly see that this is a big ethical issue when combined with the deontological perspective.

4.2 Ethical Issue 2: Graphite Tip Control Rods Catalyst

Ethical Issue

In order to save costs, Soviet engineers used graphite tip boron control rods in the reactor chamber. Graphite is a positive void coefficient material and creates that feedback loop we talked about in the background. It is the catalyst of the explosion when combined with the already positive void coefficient reactor design which leads to the ultimate catastrophe scenario.

Deontological Analysis

From a duties agent based perspective, it is still dubious as to what the engineers could have done. As humans their first duty is to their family, in the environment they were in they could not have argued about not cutting costs and going against the wishes of the higher ups to save money. Because of their life being potentially voided when they try that I could see the engineers saying that the chance for catastrophe happening is low enough that they were willing

to keep on with it. From a patient centered view though this is just an abhorrent decision given that you are willingly setting up a situation for potential failure if a set of wrong decisions happen, the patients have the right to not be set up in a potentially life threatening situation when they have 0 autonomy in that situation.

Reflexive Principlism Analysis

Looking at it from the reflexive framework a lot of the analysis remains the same from the last one. Beneficence is decent since it gives the stakeholders power while also keeping budget low, however the non-maleficence is all the way down in the gutter to the point where we don't even need to look at any of the other points since it will just overpower everything else. Similarly though for Justice there is nothing really it is neutral and for autonomy it is again a decision that provides 0 autonomy to the stakeholders.

5. Options for Resolution

Option A: Argue and Try to Corrupt or Deceive Higher Ups

Option B: Institute Better Failsafes

Option C: Whistleblow

6. Recommendation : Argue and Try to Corrupt or Deceive Higher Ups

One could argue that instituting better failsafes could be a way to fall in line with agent rights and duties along with maximizing beneficence and non maleficence however the chance that the higher ups would allow that seems close to zero since they already wanted to cut costs with the graphite rods. Similarly, whistleblowing is simply not even close to an option since it would in almost 999 out of 1000 cases lead to just immediate death or death in a few years if they could find an American embassy. Given the situation the designers were in, the only option they had was to lie to the Higher ups. While in some Kantian perspectives lying is something one should never do. Thankfully we are not Kantian deontologists and as such when we use a combination of agent patient deontology along with reflexive principlism we can see that simply lying to the higher ups and saying that graphite would not work as a boron substitute or that positive void coefficients would not work to implement in a reactor of this size or with this budget. This would be aligned with the duties designers had to their family, their rights to the public along with however being slightly less beneficent but increasing in non maleficence by a lot more than the loss in beneficence. All things considered this would provide the most ethically sound decisions to the reactor designers. This ultimately not only provides the most ethically sound decision but also saves the day by defusing the situation, no pun intended.